

THE UNIVERSITY OF ARIZONA
GRADUATE COLLEGE

As members of the Dissertation Committee, we certify that we have read the dissertation prepared by: *Martin E. Liza*, titled: *Aero-Optical Effects in High-Enthalpy Flows* and recommend that it be accepted as fulfilling the dissertation requirement for the Degree of Doctor of Philosophy.

Kyle Hanquist

Date: 10/02/2025

Kyle M. Hanquist

Wayne Hacker

Wayne Hacker (Oct 3, 2025 13:53:23 PDT)

Date: 10/03/2025

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Jerzy W. Rozenblit

Date: 10/05/2025

Jerzy W. Rozenblit

Final approval and acceptance of this dissertation is contingent upon the candidate's submission of the final copies of the dissertation to the Graduate College.

I hereby certify that I have read this dissertation prepared under my direction and recommend that it be accepted as fulfilling the dissertation requirement.

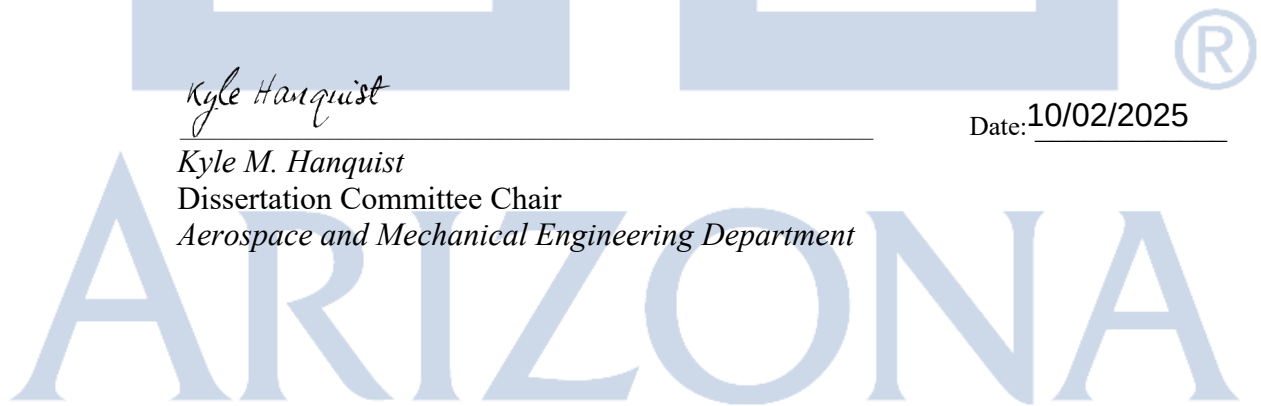
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